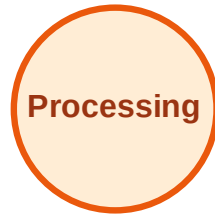


BFS Traversal

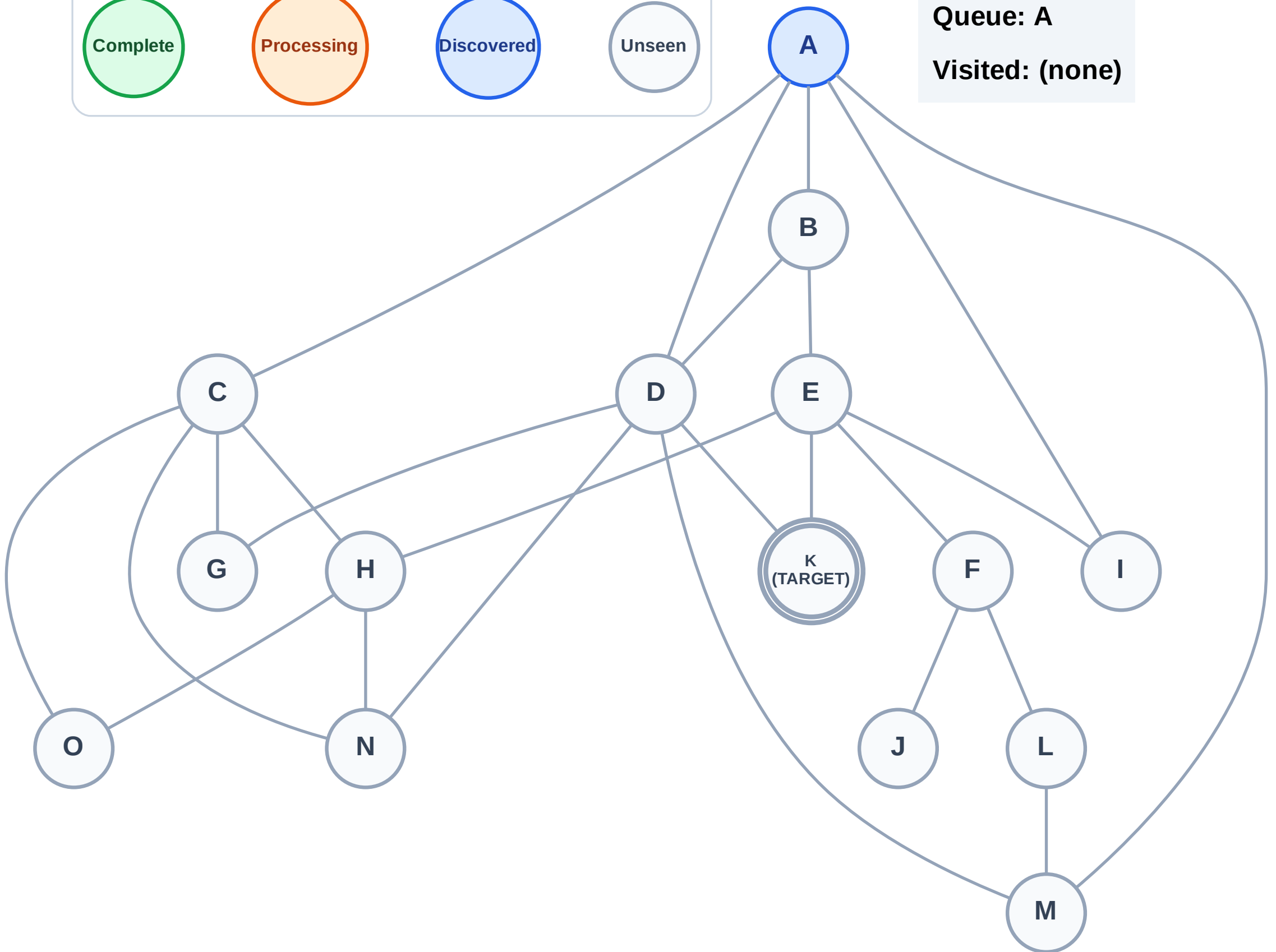
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

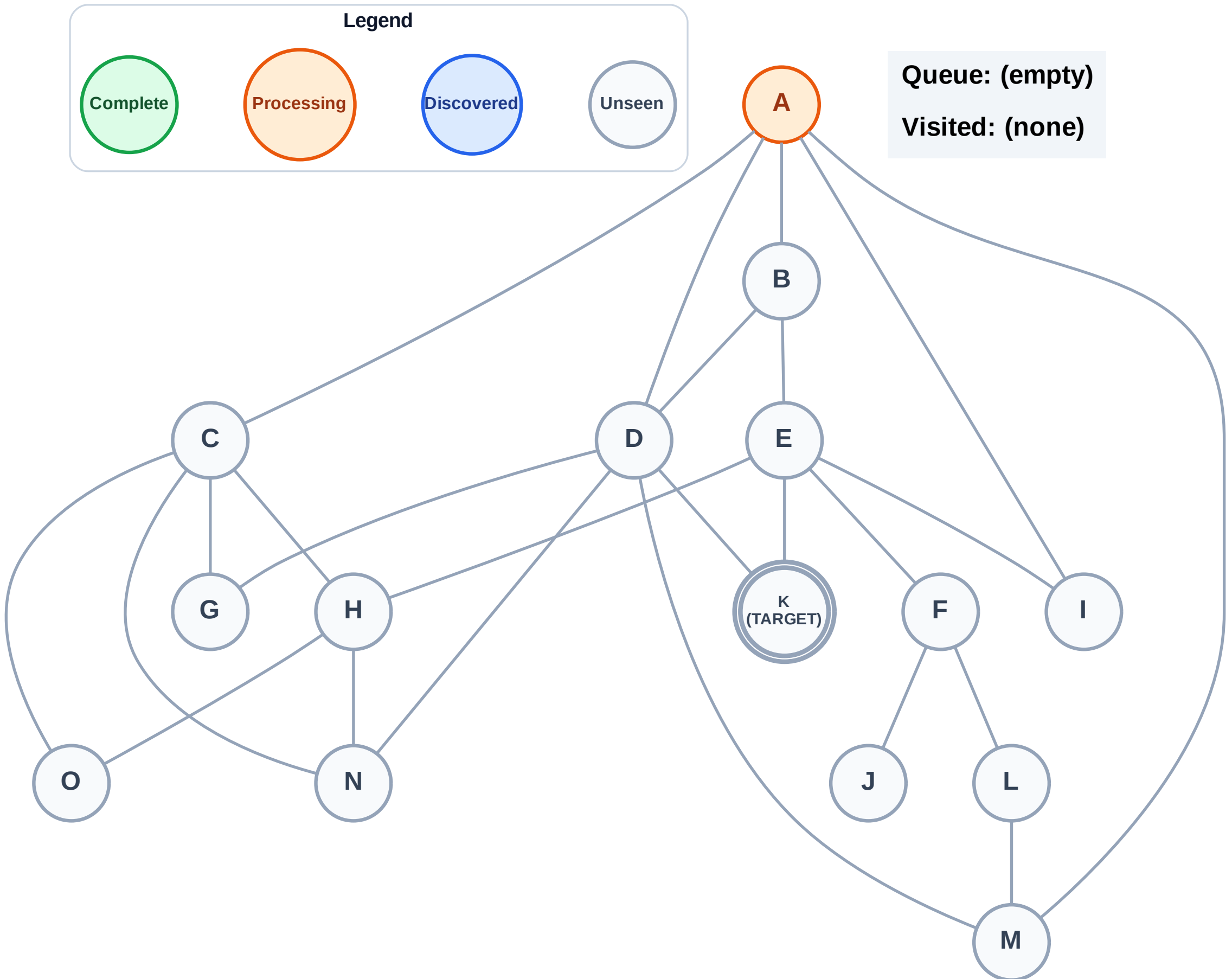


Queue: A

Visited: (none)



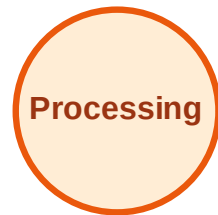
Step 2: Process node A



BFS Traversal

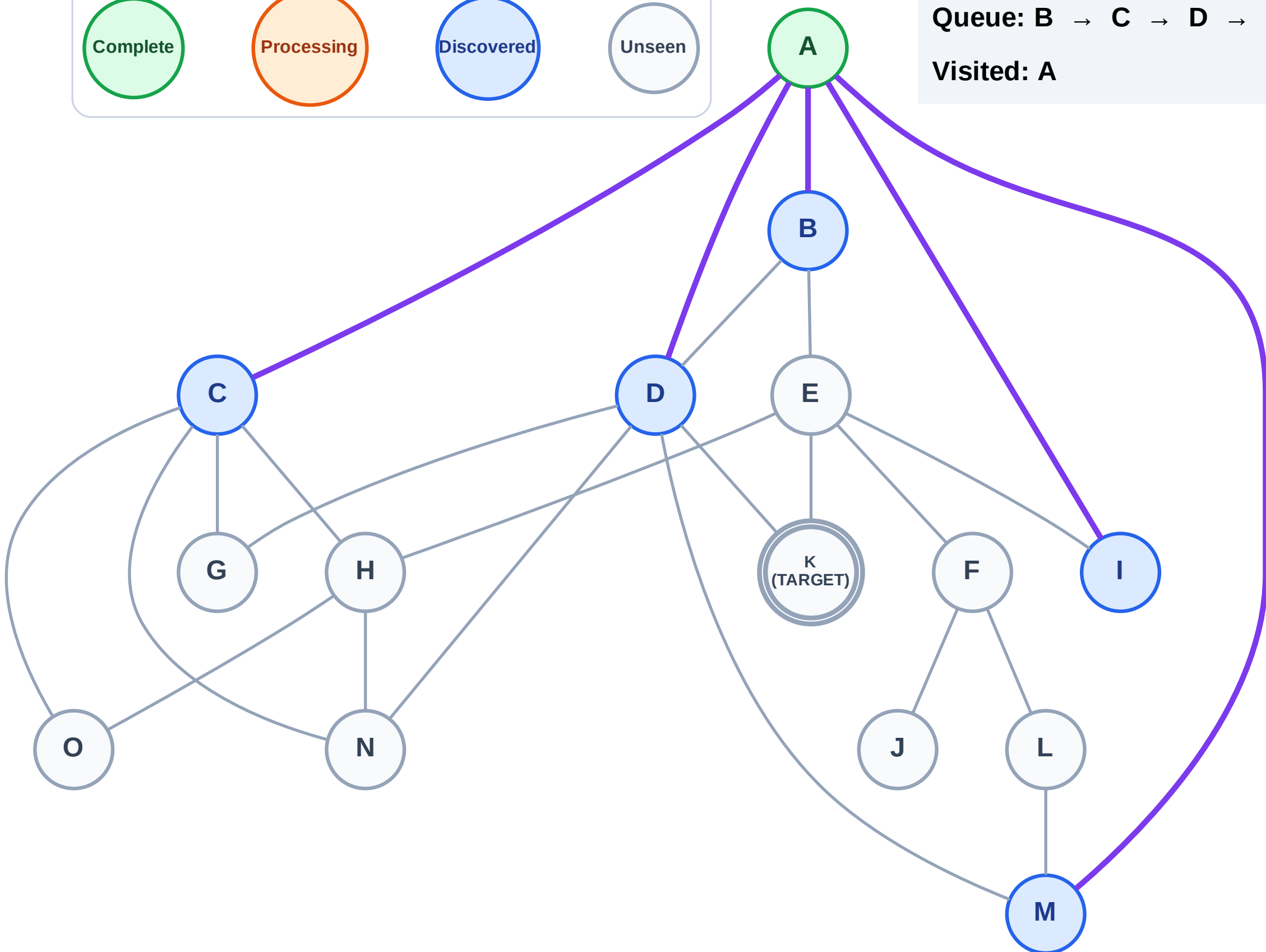
Step 3: Discover B, C, D, I, M from A

Legend



Queue: B → C → D → I → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

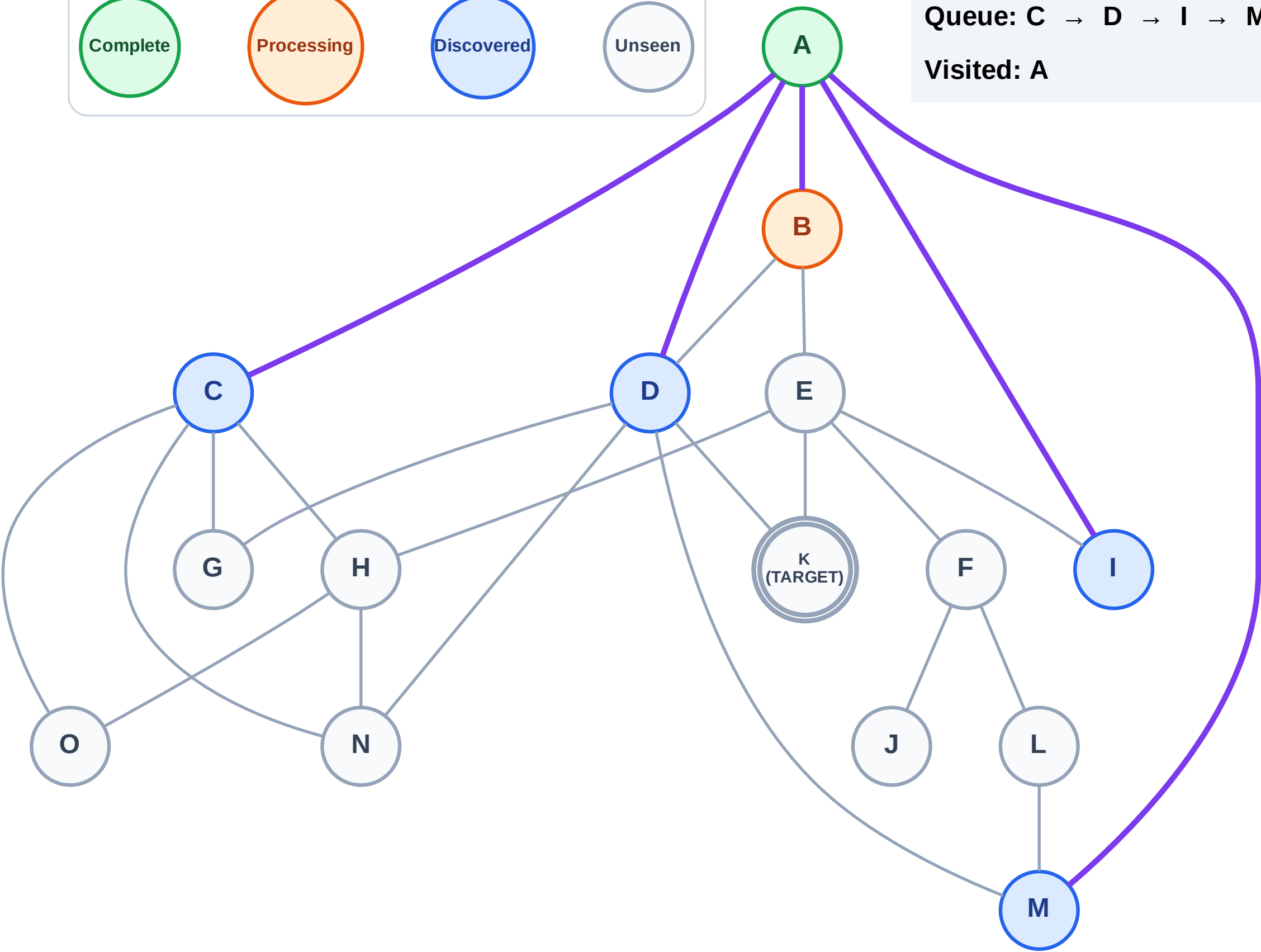
Processing

Discovered

Unseen

Queue: C → D → I → M

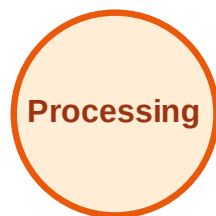
Visited: A



BFS Traversal

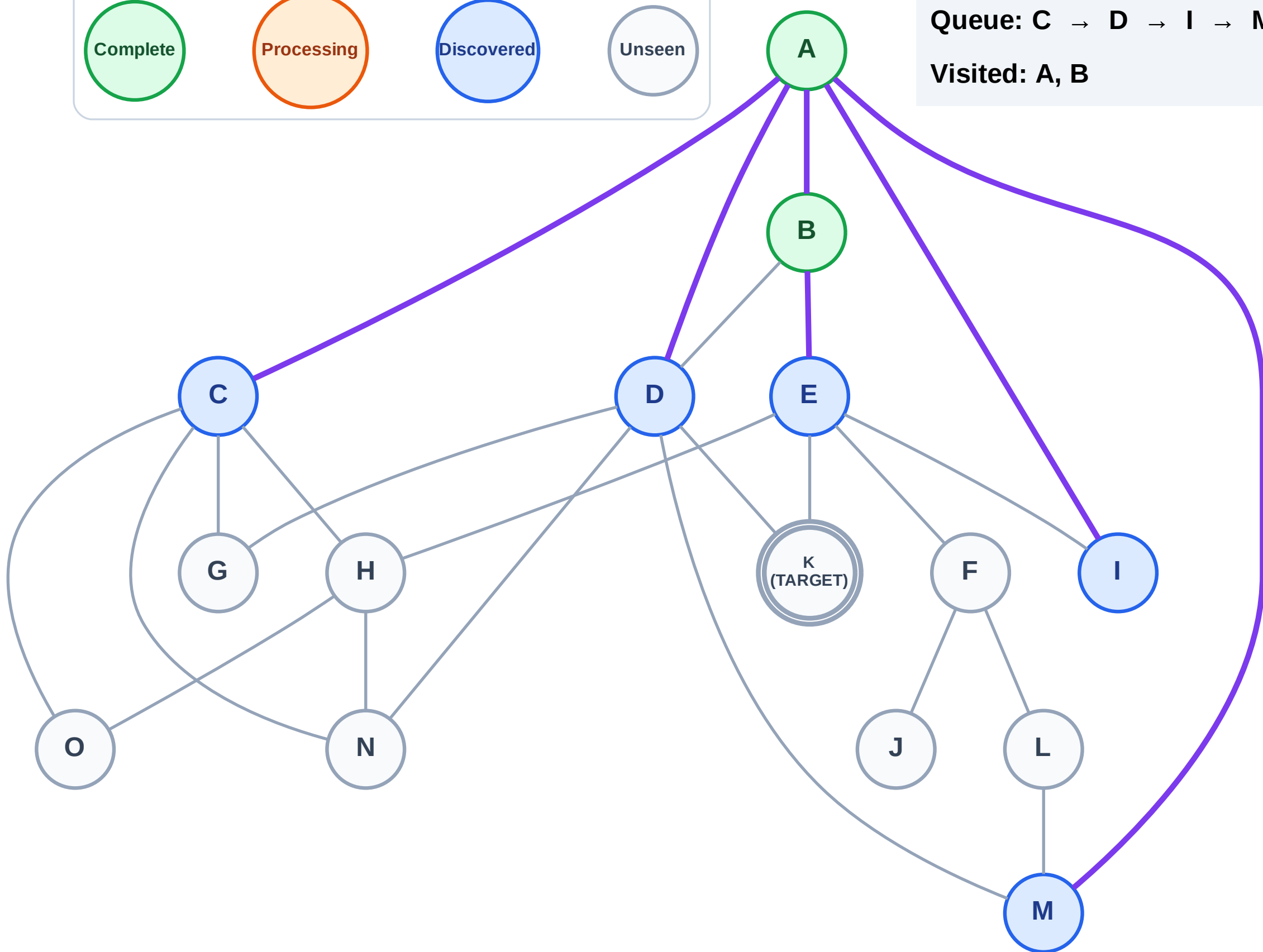
Step 5: Discover E from B

Legend



Queue: C → D → I → M → E

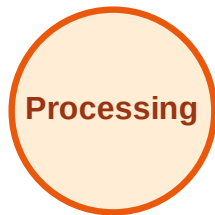
Visited: A, B



BFS Traversal

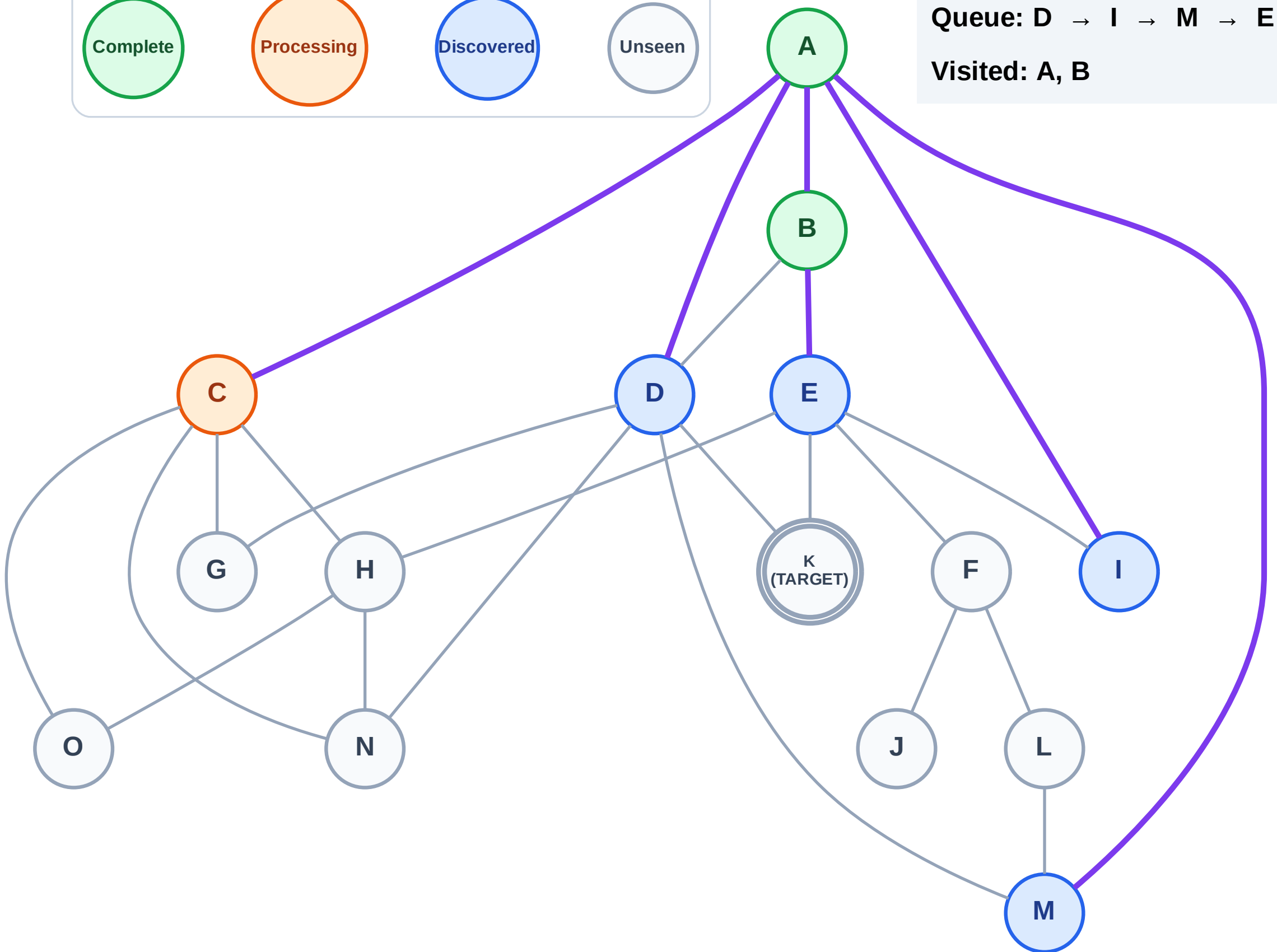
Step 6: Process node C

Legend



Queue: D → I → M → E

Visited: A, B



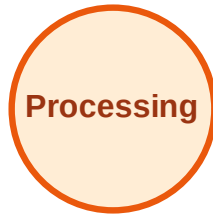
BFS Traversal

Step 7: Discover G, H, N, O from C

Legend



Complete



Processing



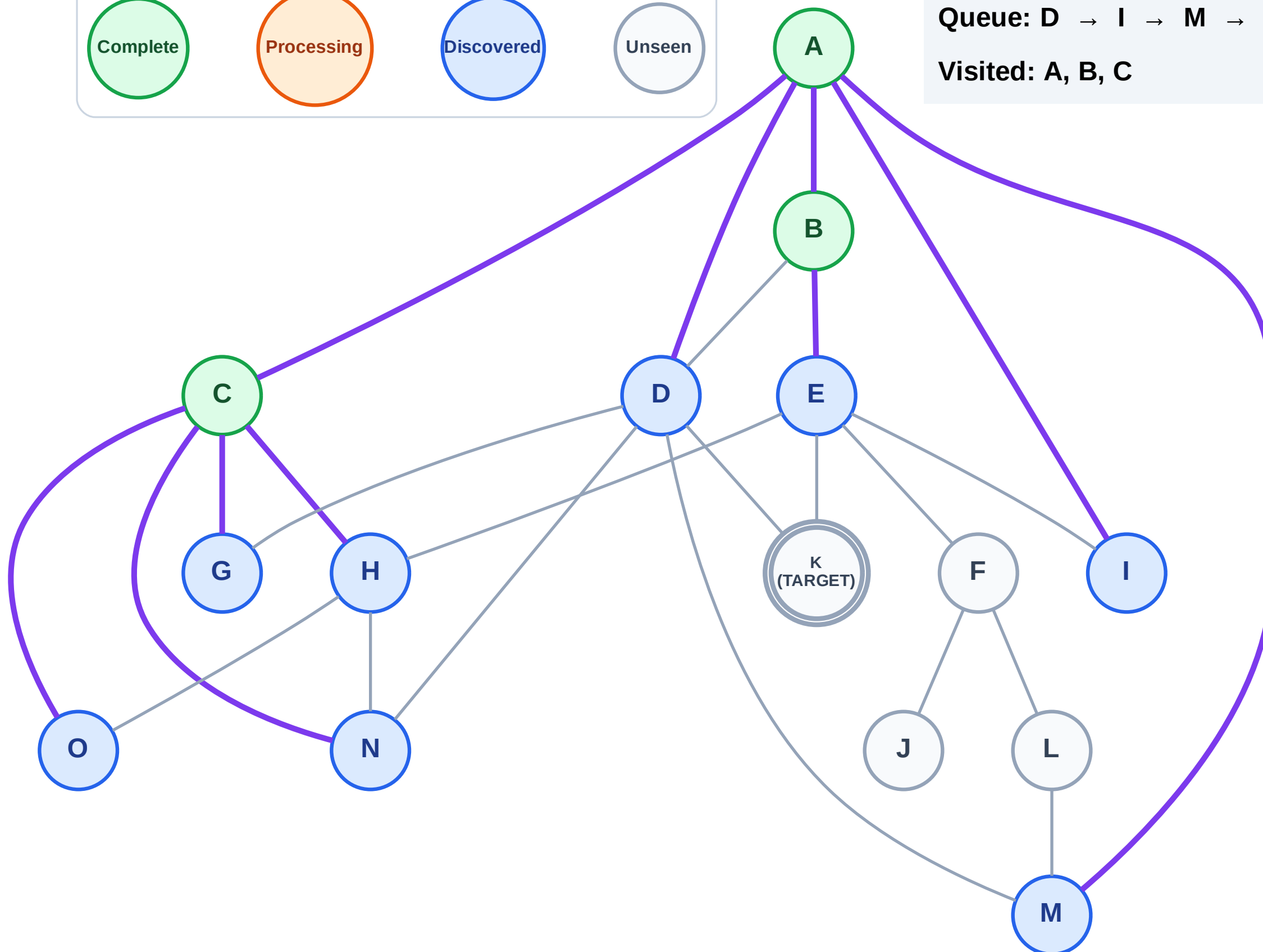
Discovered



Unseen

Queue: D → I → M → E → G → H → N → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

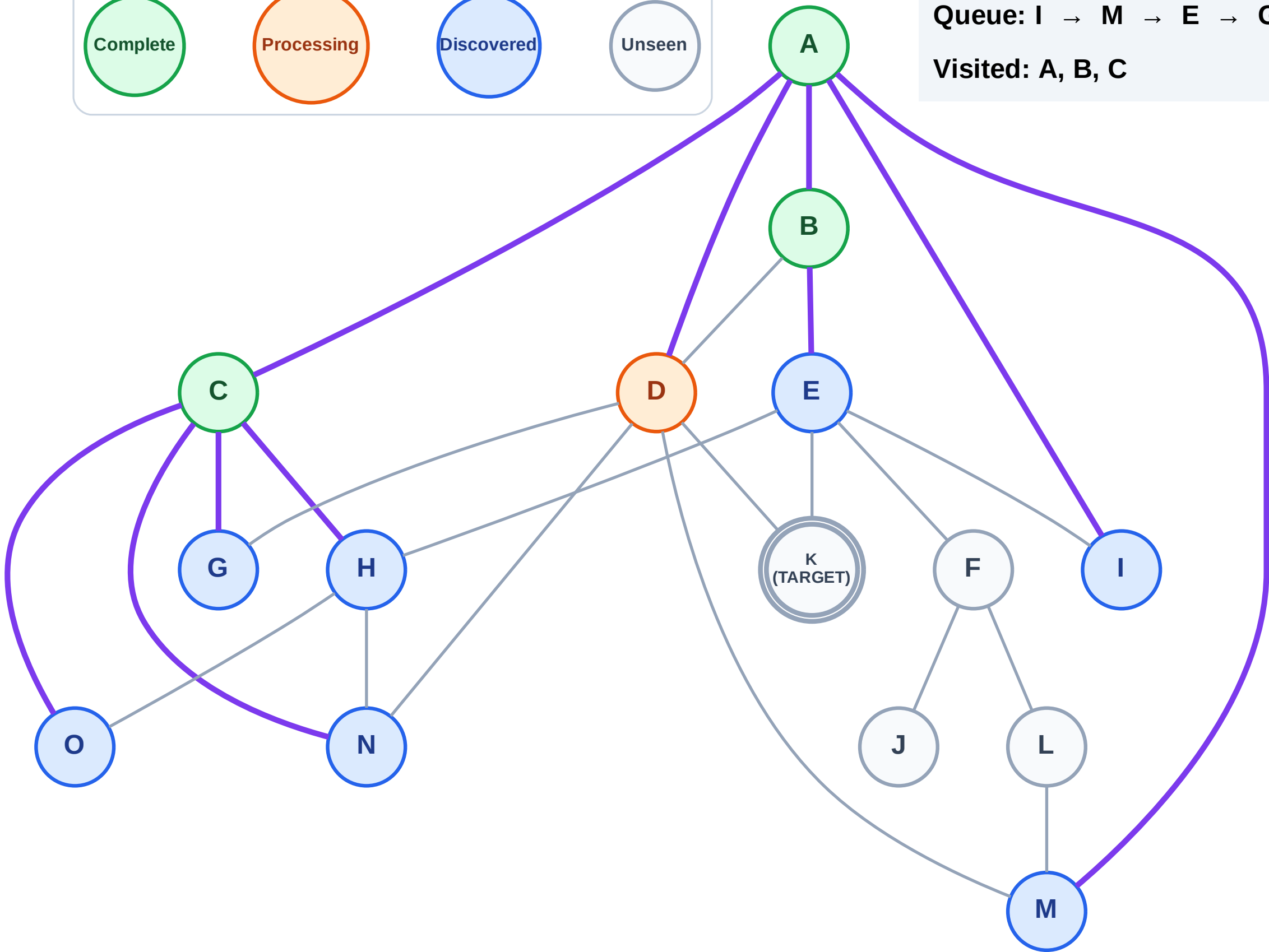
Processing

Discovered

Unseen

Queue: I → M → E → G → H → N → O

Visited: A, B, C



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

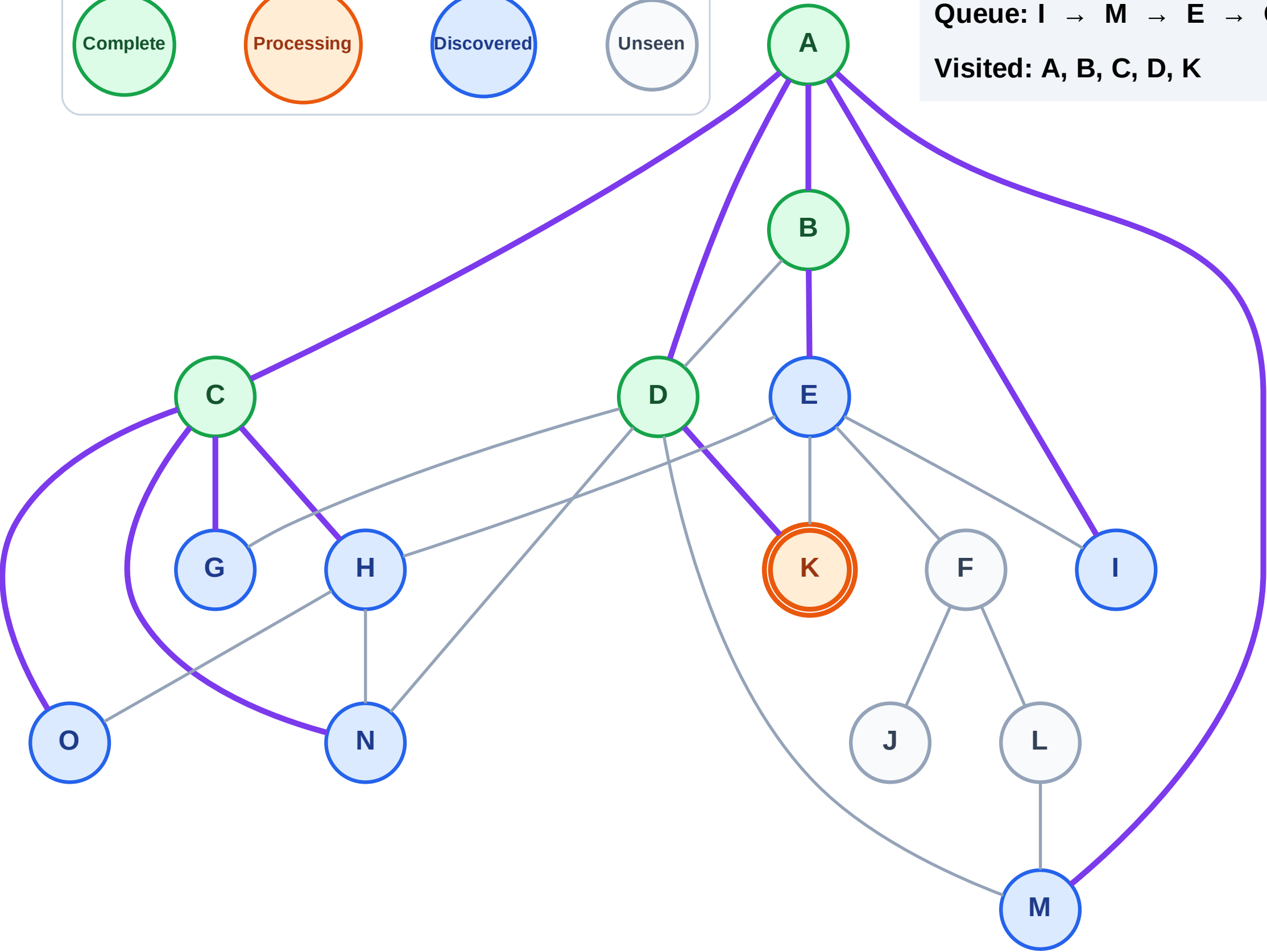
Processing

Discovered

Unseen

Queue: I → M → E → G → H → N → O

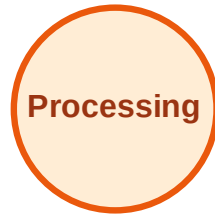
Visited: A, B, C, D, K



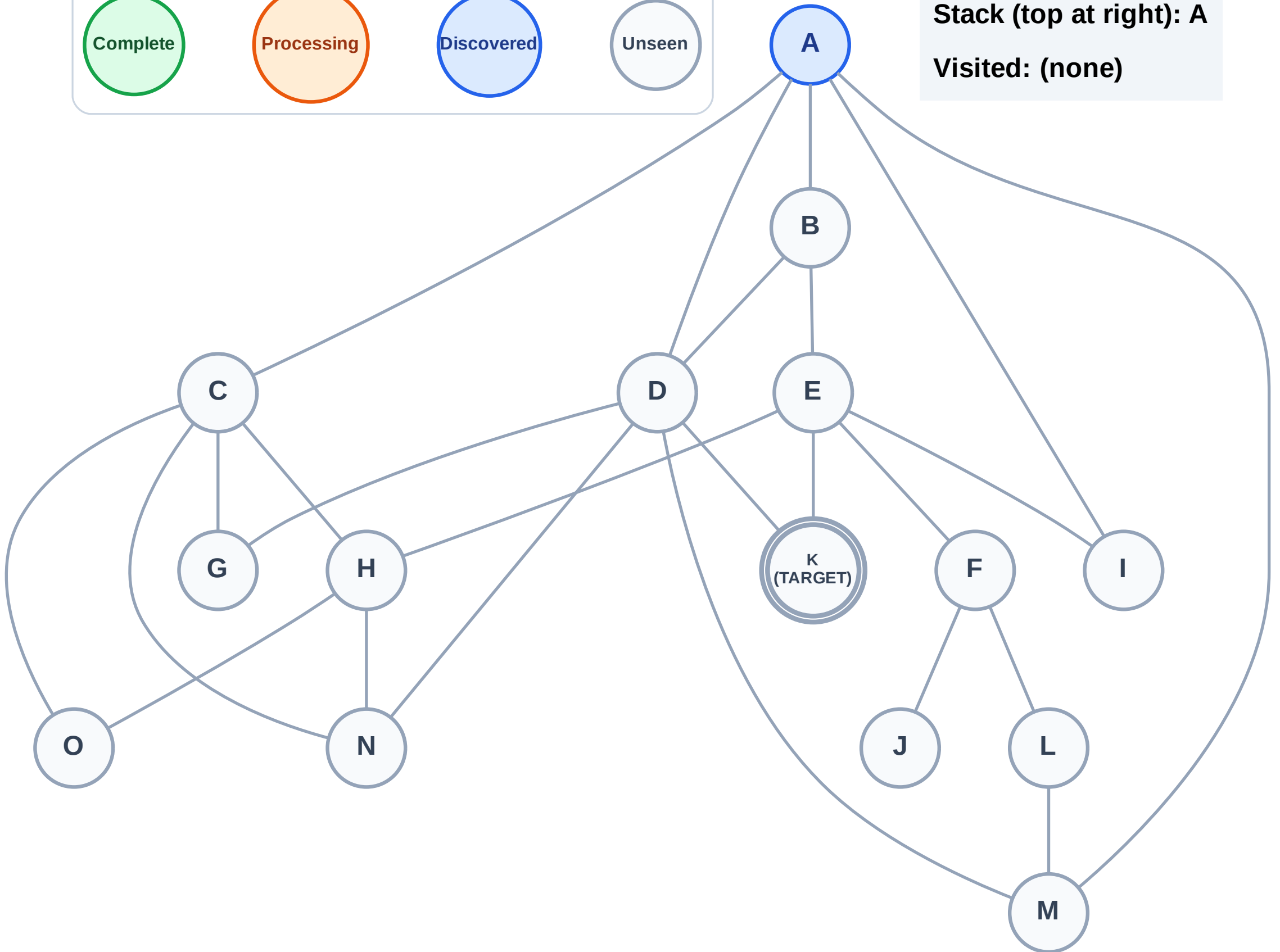
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

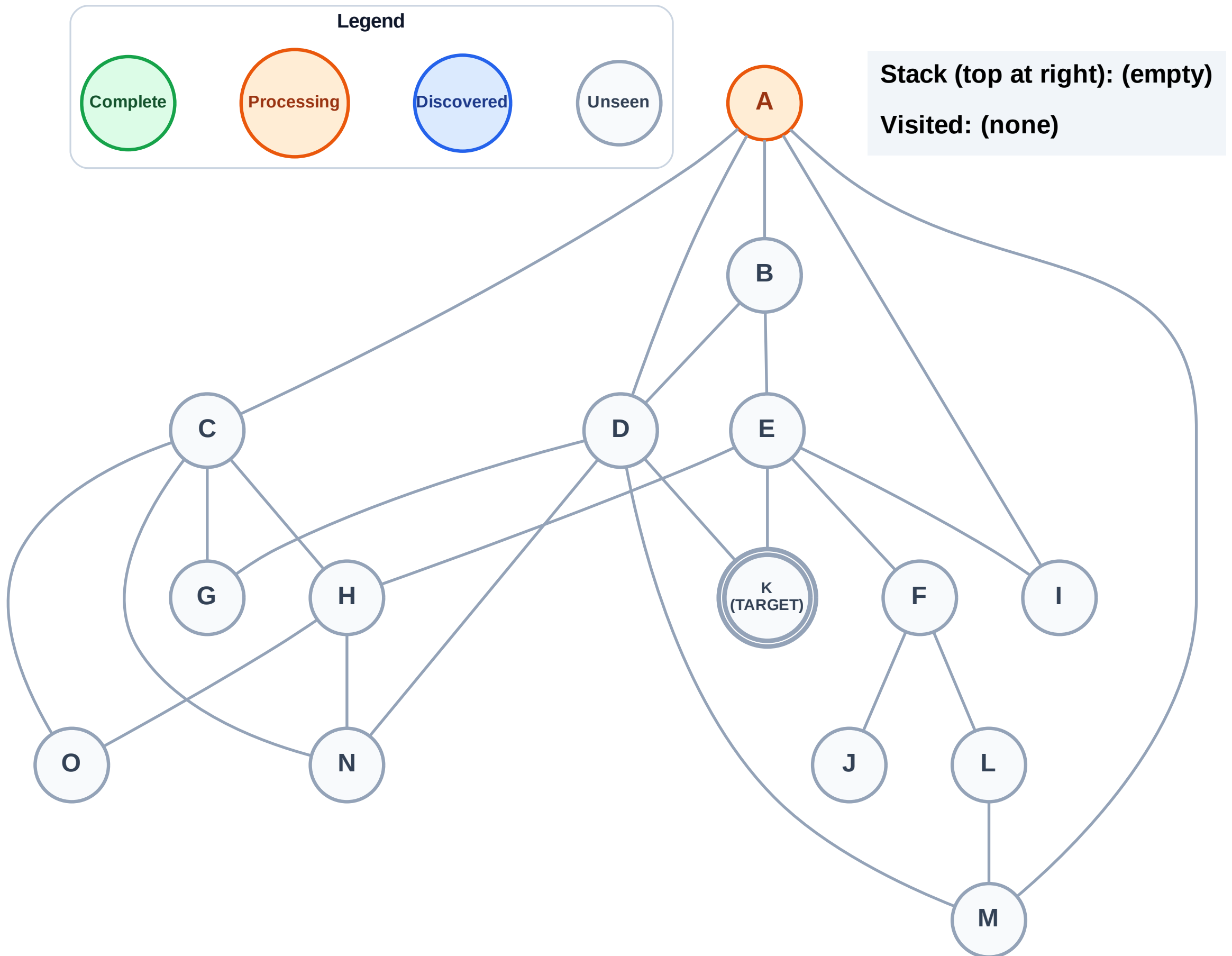
Legend



Stack (top at right): A
Visited: (none)



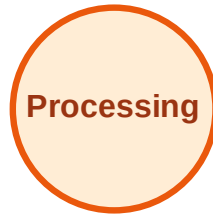
Step 2: Process node A



DFS Traversal

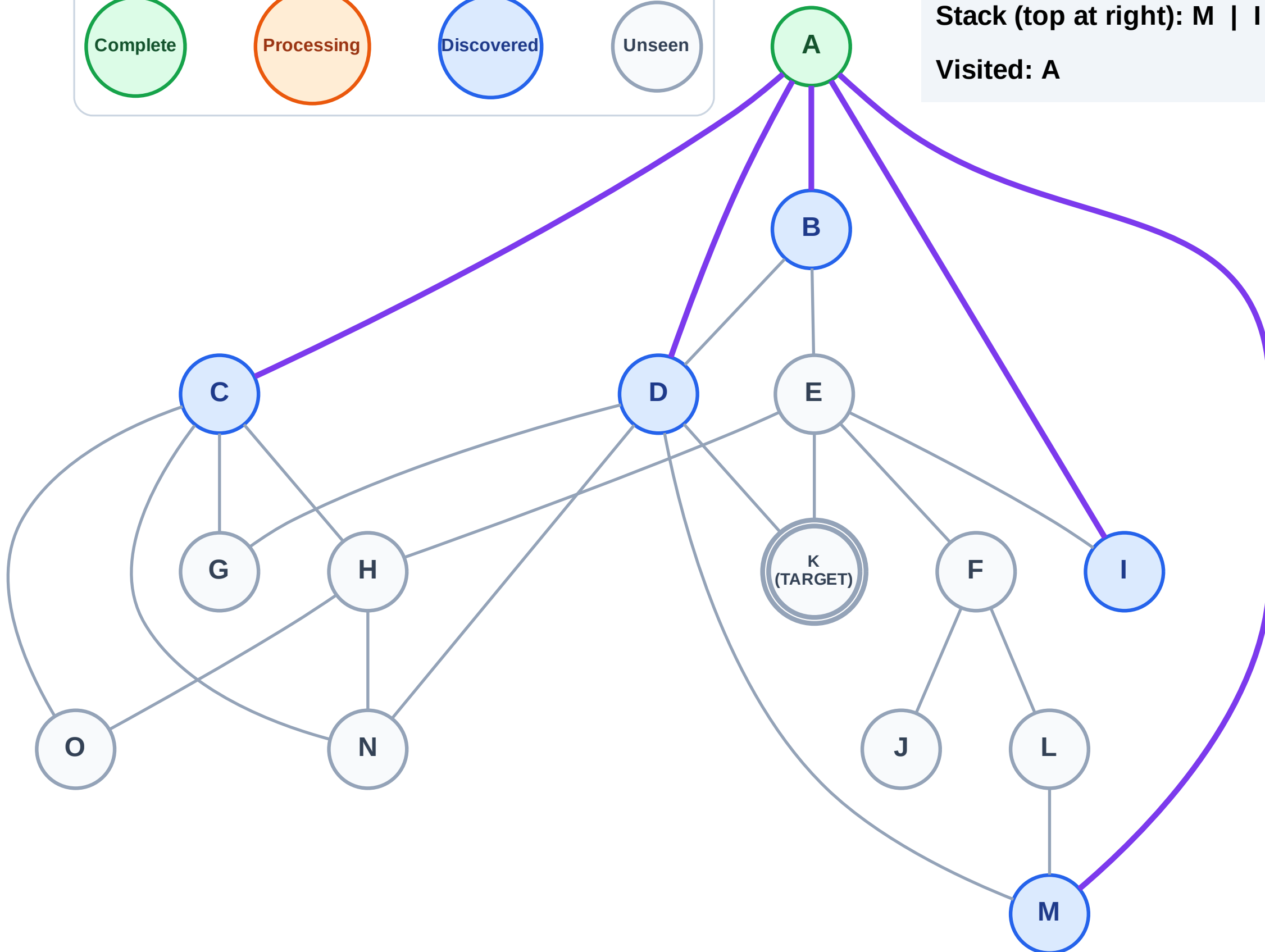
Step 3: Discover M, I, D, C, B from A

Legend



Stack (top at right): M | I | D | C | B

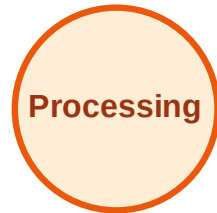
Visited: A



DFS Traversal

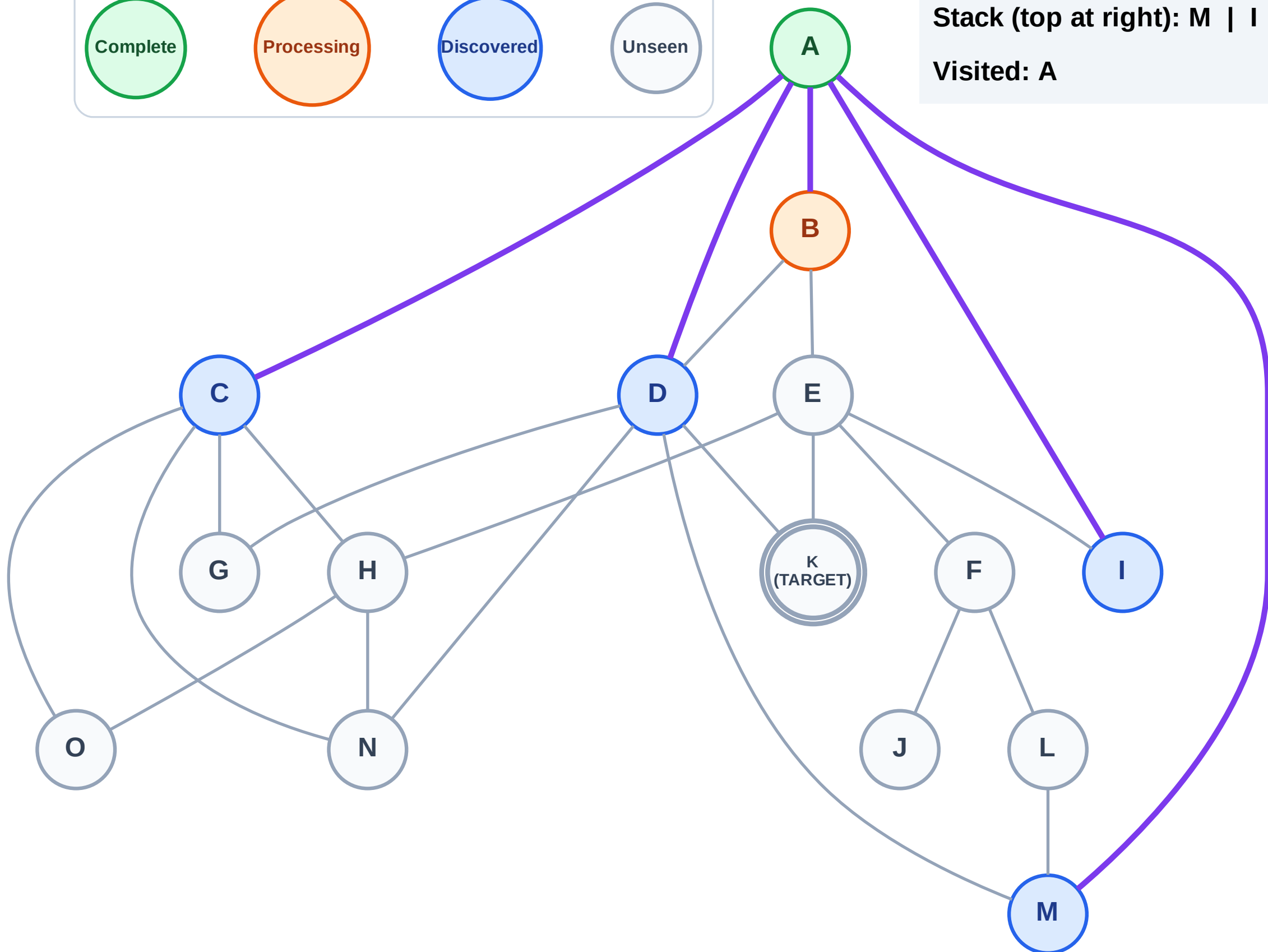
Step 4: Process node B

Legend



Stack (top at right): M | I | D | C

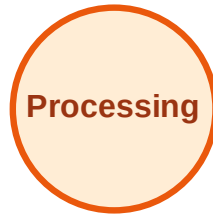
Visited: A



DFS Traversal

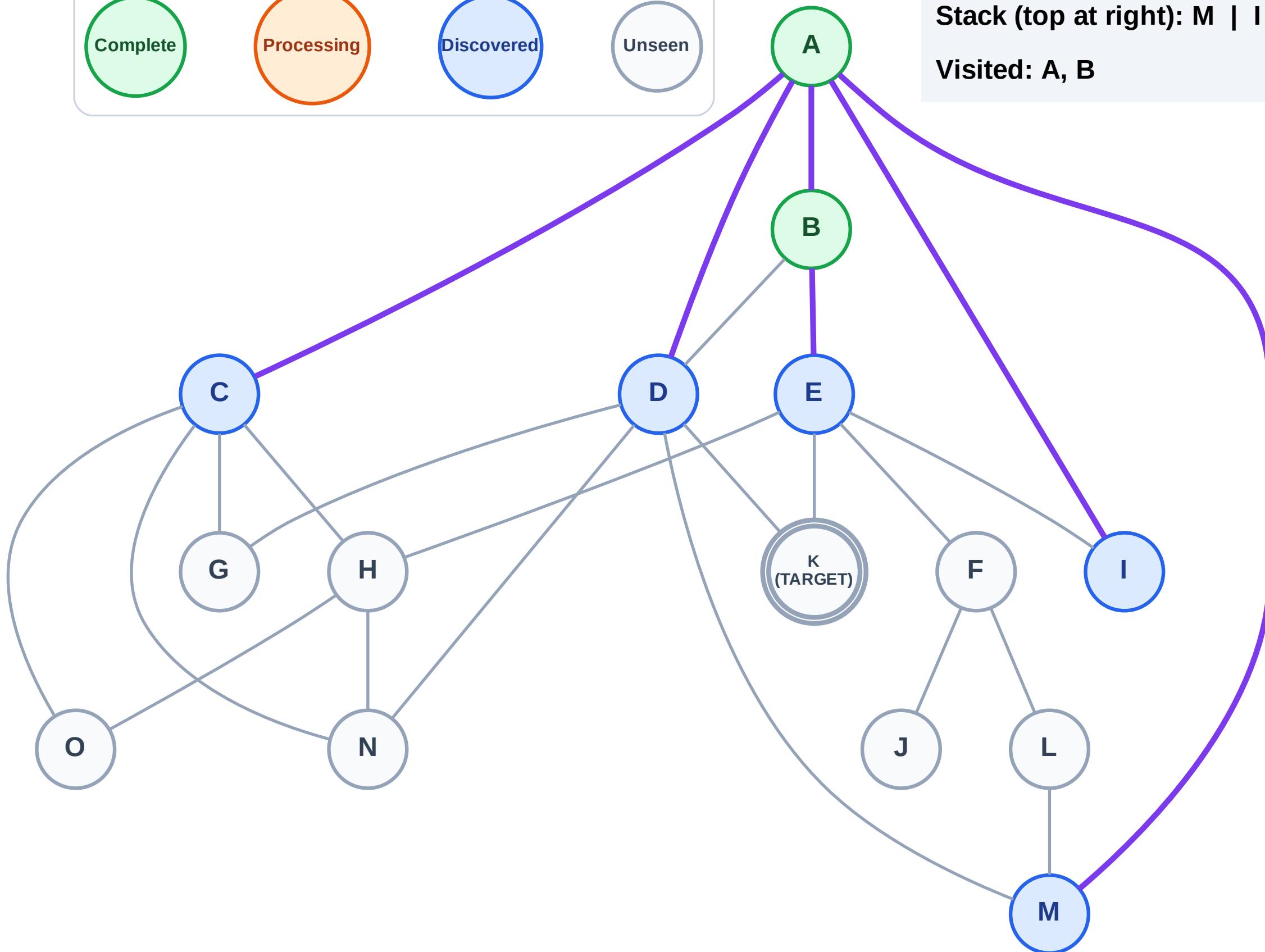
Step 5: Discover E from B

Legend



Stack (top at right): M | I | D | C | E

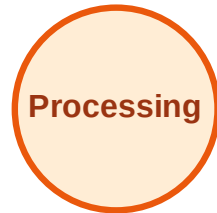
Visited: A, B



DFS Traversal

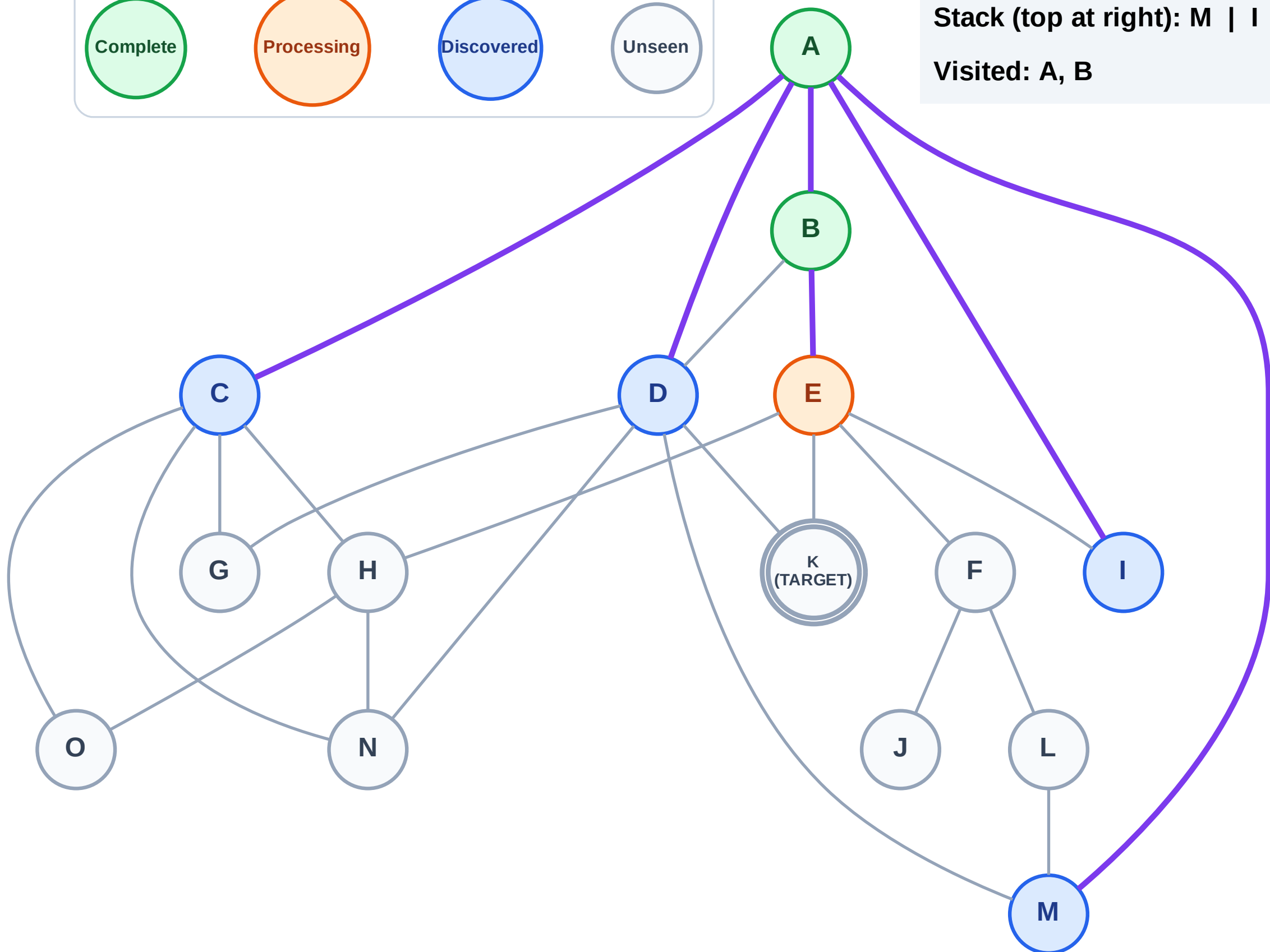
Step 6: Process node E

Legend



Stack (top at right): M | I | D | C

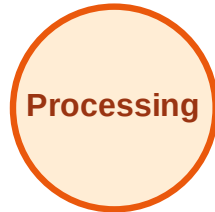
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend



Stack (top at right): M | I | D | C

Visited: A, B, E, K

